



East West University

Department of Computer Science and Engineering

Semester : Spring 2021

Course Title: Electrical Circuits

Course Code: CSE209

Section: 03

Experiment Number: 07

Experiment Title: DC Circuit Analysis in PSpice using
Source and Resistance Sweep

Student Name: Md. Rakibur Rahman

ID: 2019-3-60-113

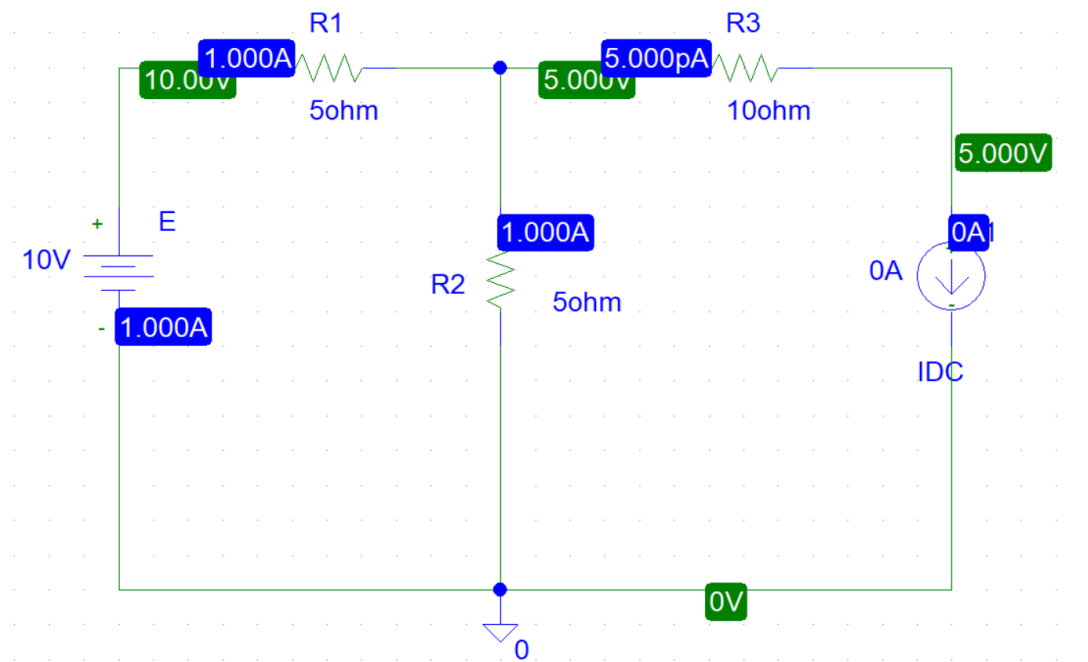
Date of Performance: 05.05.2021

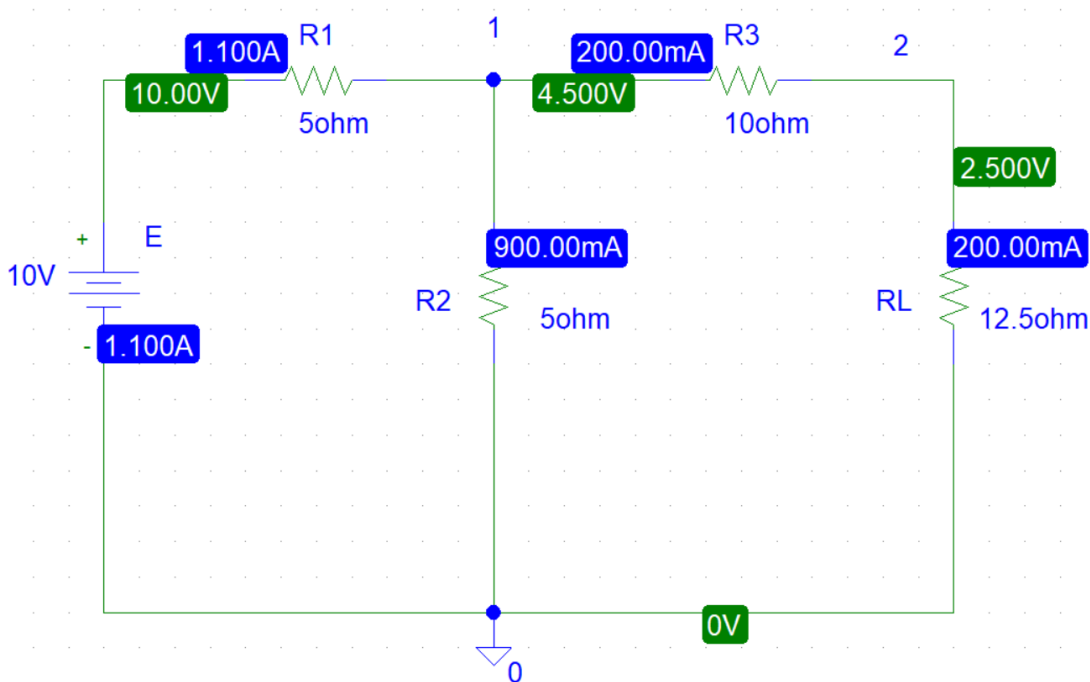
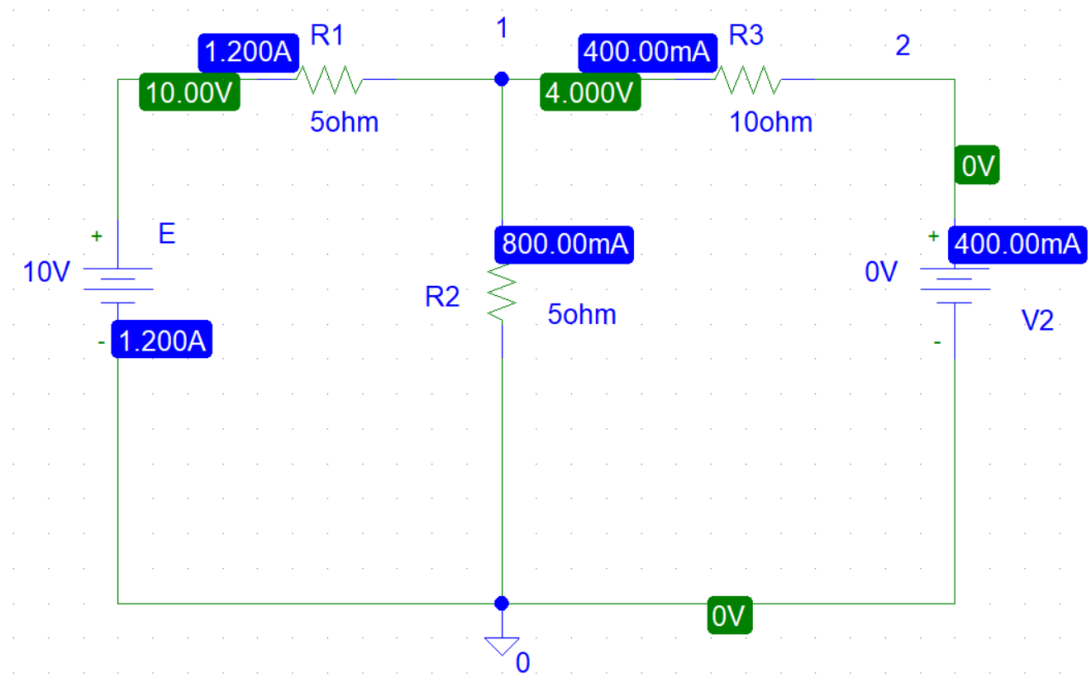
Date of Submission: 09.05.2021

Objectives:

1. To analyze DC circuit in PSpice by sweeping source and resistance.
2. To verify maximum power transfer theorem.

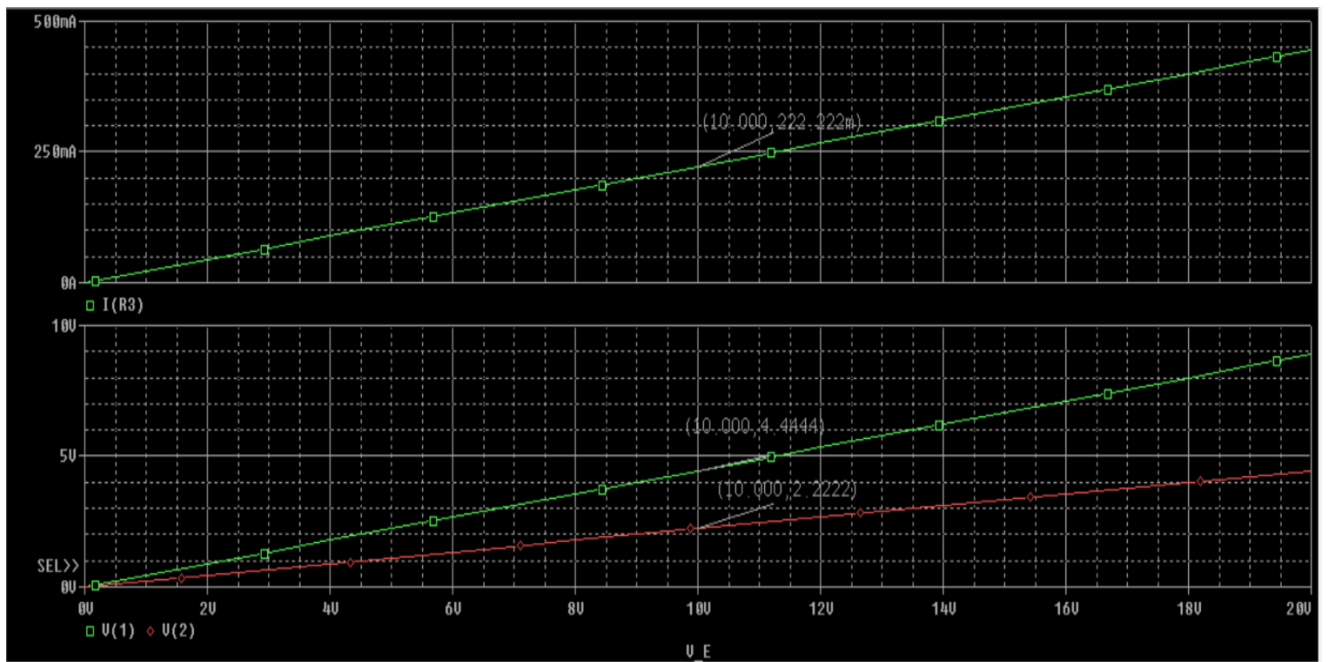
Circuit Diagram:



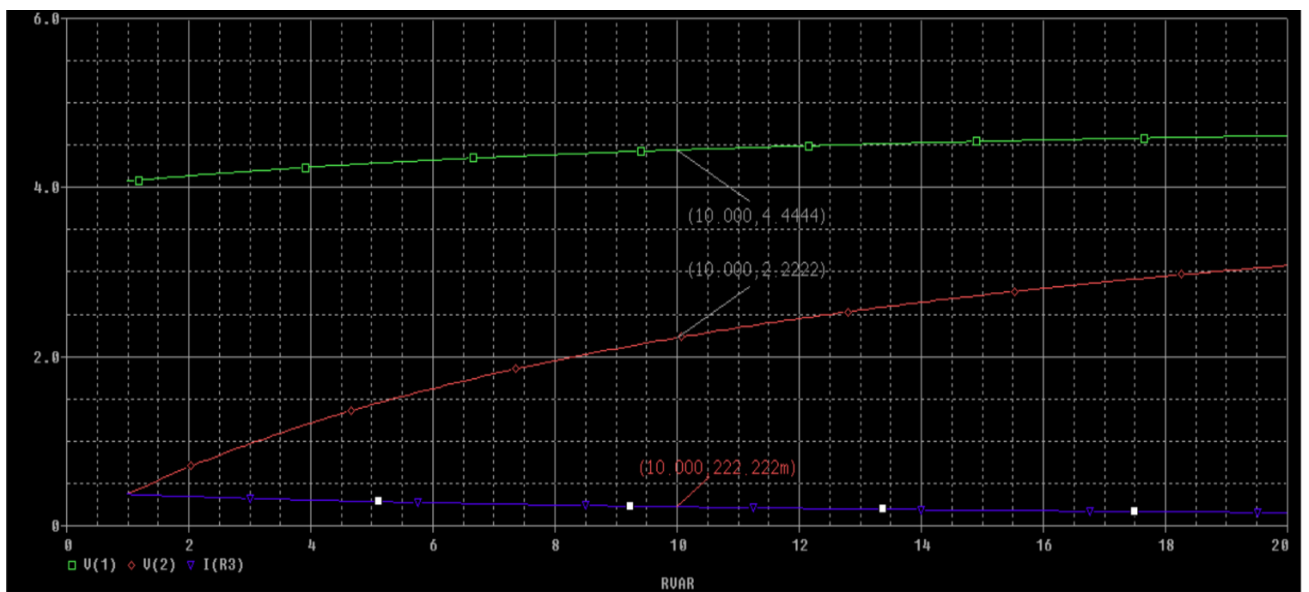


Answers to the post lab report questions:

1. From step 4,



From step 5(d),

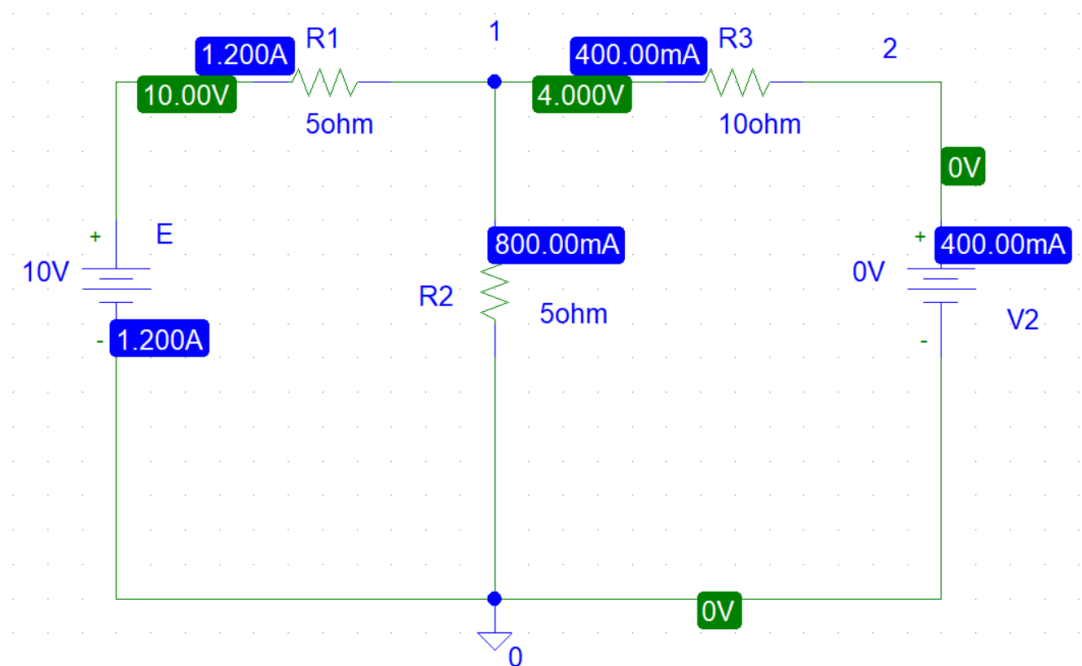


Comparing the values of $V(1)$, $V(2)$ and $I(R_3)$ obtained in steps 4 and 5(d),

Values	Step 4	Step 5(d)
V_1	4.4444V	4.4444V
V_2	2.2222V	2.2222V
$I(R_3)$	222.222mA	222.222mA

Here, the values of step 4 & step 5(d) are same.

1. From step 2,



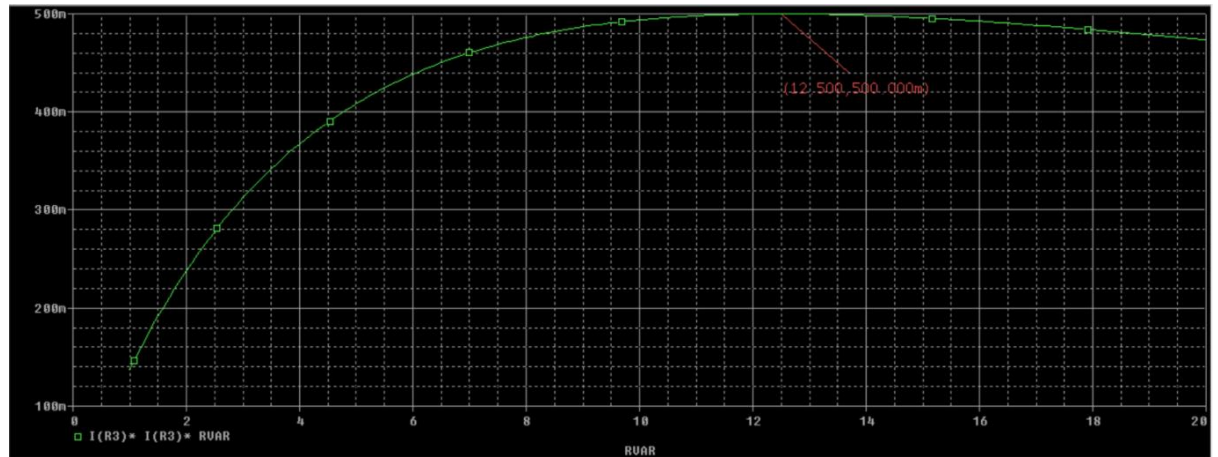
Here , $I_{SC} = 400\text{mA} = 0.4\text{A}$

$V_{th} = 5\text{V}$ [from step 1]

$$R_{th} = \frac{V_{th}}{I_{SC}} = \frac{5}{0.4} \Omega = 12.5 \Omega$$

Here , $R_{th} = R_L = 12.5$

From step 5(e),

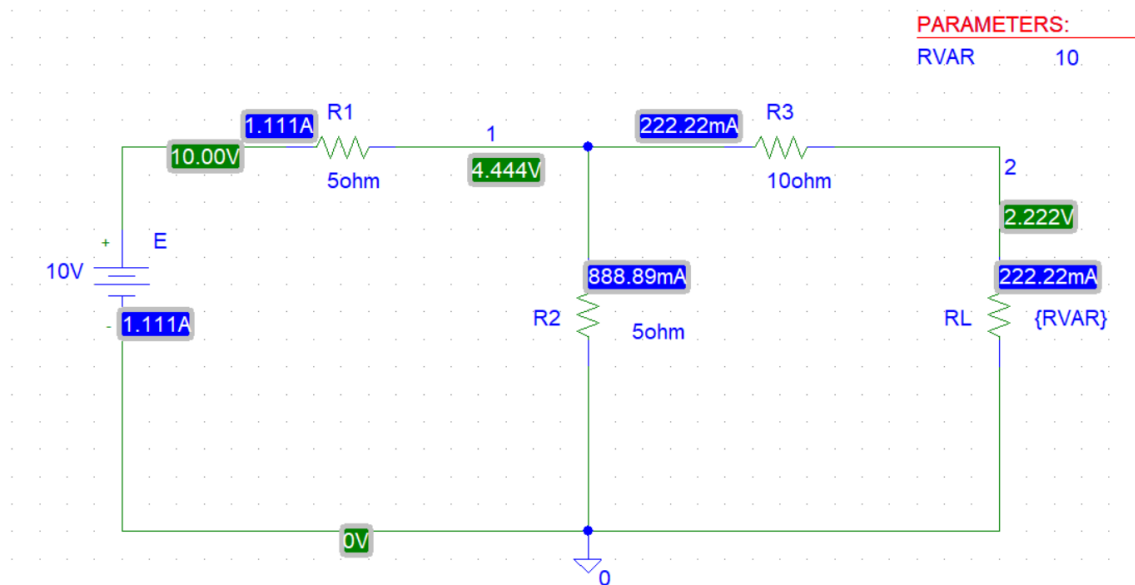


Comparing the load resistance R_L for maximum power transfer obtained in steps 2 and 5(e),

Step 2	Step 5(e)
12.5 Ω	12.5 Ω

R_L for maximum power transfer obtained in step 2 & 5(e) are same.

3.



Applying KCL at node 1,

$$\frac{V_1 - 10}{5} + \frac{V_1 - V_2}{10} + \frac{V_1}{5} = 0$$

$$0.5V_1 - 0.1V_2 = 2 \dots (1)$$

Applying KCL at node 2,

$$\frac{V_2 - V_1}{10} + \frac{V_2}{10} = 0$$

$$-0.1V_1 + 0.2V_2 = 0 \dots (2)$$

Solving equation 1 & 2,

$$V_1 = 4.4444 \text{ V}$$

$$V_2 = 2.2222 \text{ V}$$

$$I(R_3) = \frac{V_1 - V_2}{R_3}$$

$$= \frac{4.4444 - 2.2222}{10}$$

$$= 222.22 \text{mA}$$

Comparing the theoretical solutions with the solutions obtained from PSpice,

Values	PSpice	Calculated
R_L	12.5 Ω	12.5 Ω
V_1	4.444V	4.4444V
V_2	2.222V	2.2222V
$I(R_3)$	222.22mA	222.22mA

The calculated values & the simulated values are the same. Thus, there is no discrepancy.

Conclusion: In this experiment , we learnt about analyzing DC circuit in PSpice by sweeping source and resistance. And verifying maximum power transfer theorem. We found that, there is no discrepancy in the calculated & simulated values.